

EXAMUDAAN — GOVERNMENT EXAM & RECRUITMENT AGGREGATOR — EXECUTIVE ARCHITECTURE

High-Availability Ingestion. Zero-Lag 2-Duplicate Guard. Dual-Language AI Engine (English & Marathi). Real-Time Aspirant Alerting.

USERS & CHANNELS

Aspirants / Students
(All-India & State Jobs)

Mobile Job Seekers
(Responsive Drawer & Nav)

Desktop Applicants
(Sticky Sidebar Filters)

Admins & Editorial Ops
(Feedback & Suggestion Desk)

Public Web Portals
(Next.js 14 App Router)

Interactive AI Matcher
(AiMatcher.js Eligibility)

Paid Alert Subscribers
(WhatsApp / SMS ₹29/49/99)

SECURITY LAYER

Data Encryption
(In Transit TLS 1.3
& At Rest AES-256)

IAM & Secrets
(Rotated Gemini Keys
& PostgreSQL .env)

Session & Auth
(Custom JWT / lib/auth.js
HttpOnly Cookies)

Anti-Ban / Throttling
(Rotating User-Agents
& Crawl Delay 1-3s)

DPDP Act 2023
(Gov Non-Affiliation
& User Data Consent)

Audit & Traceability
(Monthly Partitioned
Raw HTML & Logs)

API

API & INGESTION GATEWAY
Scrapy Crawler Daemon, Scheduled Runner, Next.js REST API & Route Dispatcher

REST APIs: /api/notifications (Universal filtered listing) | /api/feedback (Suggestions & Bug desk)
Ingestion Control: run_scraper.py CLI | cron_6h.sh (6-hour cron scheduler) | Async Pipeline Dispatcher
Query Mapping: ?type=job → recruitment | ?type=result → result | ?type=admit-card → admit_card

SCRAPE

SCRAPING & CRAWLING ENGINE
34+ Autonomous Portals, Dynamic JS Headless Crawler & 2-Tier Deduplication

Central Portals: SSC, UPSC, RRB, IBPS, SBI, NTA, BPSC, Employment News (Every 6h)
Maharashtra & Municipal: MPSC (Crawl4AI Headless Browser), MahaPolice, Mumbai/Thane Police, BMC, PMC, TMC, SRPF, Prisons
Research & PSUs: ACTREC, AIIMS Nagpur, CSIR-NEERI, IIPS, IISER Pune, ICT, CIRCOT, NBSSLUP, Konkan Railway, MECL, MOIL
2-Tier Deduplication: DuplicateStopMixin (Early CloseSpider after 2 dups) + DeduplicationPipeline (Priority 10)

AI

AI / ML & INTELLIGENCE LAYER
Multi-LLM Native PDF Extraction, Pydantic Structured Output & Marathi Translation

Official PDF Analysis: Google Gemini 2.5 Flash native PDF reader with multi-key rotation (DeepSeek-V3 & Groq fallbacks)
Promoted Fast-Filters: Auto-extracts total vacancies, salary range (min/max), age limits, walk-in interview dates & venues
Deep Classification: ai_extracted_data JSONB (Police, Engineering, Medical, Teaching, Central vs State Govt levels)
Automatic Marathi Engine: translate_to_marathi.py (10-item batch Gemini calls: title_mr, summary_mr, qualifications_mr)

DB

DATA & STORAGE LAYER
Transactional PostgreSQL, JSONB Semi-Structured Payloads & Monthly Partitions

Primary Table: exam_notifications (1,115+ rows, indexed slug, ON CONFLICT (slug) DO UPDATE)
Granular Vacancies: posts (Per-post category, pay scale, qualifications, reservation JSONB)
Lookups & Registry: organizations (Unique UPPER acronyms: MPSC, SSC) | scrape_sources
Audit Vault: raw_scraped_data (Range-partitioned monthly by scraped_at) | scrape_log execution history

PUB

PUBLISHING & BILINGUAL ENGINE
Next.js 14 App Router, Zero-Reload English → Marathi Parity & Dynamic SEO

Universal UI Components: JobCard.js (Bilingual title & tags) | ListingPage.js (Amazon-style sticky sidebar filters)
Interactive Tools: AiMatcher.js (Instant qualification, category, age relaxation slider & live vacancy counter)
Reactive Language Engine: LanguageContext.js (Cookie/localStorage sync, instant swap to Marathi, <T> dictionary)
Automated SEO: Dynamic Meta titles, Canonical URLs, schema.org JobPosting JSON-LD for Google Jobs indexing

ALERT

NOTIFICATION & SUBSCRIPTION SERVICES
Paid Aspirant Multi-Channel Alerts, Razorpay Gateway & Admin Crawl Reports

Paid Aspirant Subscriptions: ₹29/month (Basic WhatsApp), ₹49/month (SMS + Email + PDFs), ₹99/month (VIP Custom Alerts)
Payment Processing: Razorpay Webhook & Orders API | Automated subscription activation & billing lifecycle
Scraper Health Reporting: Brevo (Sendinblue) SMTP API dispatching 6-hour crawl summaries & spider crash alerts
Multi-Channel Push (Planned): AiSensy/Gupshup WhatsApp Business API + MSG91 / Fast2SMS for instant admit card alerts

VENDOR & PARTNER INTEGRATIONS

Google Gemini AI API — gemini-2.5-flash Native PDF parsing & Marathi translation

DeepSeek & Groq Cloud APIs — LLM fallbacks for qualification extraction

Crawl4AI & Playwright — Headless browser execution for JavaScript portal rendering

Brevo (Sendinblue) SMTP API — Automated 6h scraper execution summary & error logs

Razorpay Payment Gateway — Paid subscription checkout & recurring billing webhooks

AiSensy & MSG91 — WhatsApp Business & SMS OTP gateways for candidate notifications

34+ Official Government Portals — SSC, UPSC, RRB, MPSC, BMC, PMC, State Police, etc.

PIPELINE FRAMEWORK

Standardized
Item Schemas

2-Duplicate
Early Exit Rule

Gemini Multi-Key
Rotation Engine

Rotating User-Agents
& Proxy Middleware

Idempotent DB Upsert
(ON CONFLICT)

Atomic Batch
Marathi Updates

OBSERVABILITY (SecOps | Ops | FinOps)

SECURITY & COMPLIANCE (SecOps)

- **Anti-Scraping Throttling:** 1-3s download delays, domain-specific concurrency = 1
- **Non-Affiliation Compliance:** Prominent disclaimer that ExamUdaan is a private portal
- **External Link Integrity:** Official PDF links always point to gov domains with rel= 'noopener'
- **SQL Injection Prevention:** 100% parameterized queries in pgdb.js & db.py

OPERATIONS & RELIABILITY (Ops)

- **Automated Crawl Logs:** Timestamps, duration, scraped count in scraper_YYYYMMDD_HHMMSS.log
- **Brevo Email Alerts:** Instant email summary to admin on spider success / failure
- **Graceful Failures:** Spiders catch timeouts & 404s without crashing subsequent crawlers
- **Next.js App Router Build:** All 43 static & dynamic routes validated (PASSED code 0)

COST & RESOURCE OPTIMIZATION (FinOps)

- **2-Duplicate Early Exit:** Saves 90%+ server compute and network bandwidth per crawl
- **Batch Gemini Translation:** 10 jobs per prompt cuts API costs & rate limits by 90%
- **Skip Reprocessing:** Existing records with title_mr IS NOT NULL never re-prompted
- **Pure Vanilla CSS:** Zero runtime JavaScript CSS overhead; minimal bundle size

KEY RELIABILITY METRICS: 34+ Monitored Government Portals | 1,115+ Live Database Records | 1.19s Early-Exit Latency | 43 Next.js Routes | 2 Languages (EN/MR)

CLOUD & DATA INFRASTRUCTURE (PRIMARY & VAULT RESILIENCE)

PRIMARY PRODUCTION CLOUD (Active)

Compute: Web
(Next.js 14 App Router
Vercel / Node Edge)

Compute: Scraper
(Python 3.12 VM
run_scraper.py / 6h Cron)

Primary Database
(PostgreSQL 16 Engine
10.0.2.160:5432 / test_n_new)

Headless Engine
(Playwright / Chromium
Crawl4AI for MPSC)

Connection Pools: pg (Next.js Web, 20 conns) + psycopg2-binary (Scrapy Pipeline)

STORAGE VAULT & DATA RESILIENCE (Audit & Dedup)

Raw Scraped Archive
(Monthly Partitioned
raw_scraped_data_YYYY_MM)

Preloaded Dedup Sets
(known_urls & known_hashes
in Memory at Crawl Start)

Database Backups
(Daily Point-In-Time
pg_dump Vault)

14-Day Scrape Logs
(Rotating Execution Logs
in apps/scraper/logs)

Data Replication: Direct PostgreSQL connection bypassing PgBouncer for DDL & transactional safety

Automated 6-Hour Cron Schedule
(Linux cron_6h.sh / Windows Task Scheduler)

Two-Tier 2-Duplicate Early Stop
(Closes spider in 1.2s if no new notices)

Automatic Post-Crawl Translation
(Batches of 10 notifications per Gemini prompt)

Sub-Millisecond Indexed Query Lookup
(Cached database queries via lib/pgdb.js)

KEY PRINCIPLES: High Availability (99.9%) • Real-Time Ingestion (6h Cron) • Zero-Quota Waste Deduplication • Dual-Language Parity (English & Marathi) • Strict Gov Non-Affiliation • Mobile-First Responsive UX • Cost Optimized Multi-LLM Routing

LEGEND: ■ Ingestion Gateway (Blue) ■ Scraping Engine (Purple) ■ AI / ML Intelligence (Green) ■ PostgreSQL Data (Amber) ■ Publishing Web (Teal) ■ Alerts & Subscriptions (Pink) ■ Cloud Infrastructure (Slate)